## Discrete Data Distributions

### ③ Poisson's Distribution :-

It is used to model → How many times an event occurs in a fixed interval of time.

Ex- No. of calls received in one hour.
No. of customers visiting a store in 1 day, etc.

Formula →

$$P(X=x) = \frac{e^{-\lambda} \cdot \lambda^{x}}{x!}$$

where, $\lambda$ = avg. no. of events
$x$ = event
$e$ = euler's constant

Ex- Suppose a call centre receives an avg. of 5 calls per hour. What is the probability of receiving exactly 3 calls ?

Given :- $\lambda = 5$
$x = 3$

$$P(X=3) = \frac{e^{-5} \cdot 5^{3}}{3!} = 0.14 \text{ or } 14\%$$

### ④ Multinomial Distribution :-

It is an extension to binomial distribution.

In binomial → 2 outcomes
In multinomial → more than 2 outcomes

Ex- A die, a pack of cards, etc.

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Formula :-

$$P(X=x_1 \text{ ----- } X=x_K) = \frac{n!}{x_1!\, x_2! \text{ ---- } x_K!} \cdot P_1^{x_1} P_2^{x_2} P_3^{x_3} \text{ ---- } P_K^{x_K}$$

where, $n$ = total trials
$k$ = no. of categories
$P_K$ = Prob. of each category
$x$ = event

Ques. A die is rolled 5 times. What is the probability of getting 2 fours & 3 ones.

Sol :- Given :- $n=5$
$x_1 = 2$
$x_2 = 3$
$K = 2$
$P$ = prob. $\rightsquigarrow P(X=4) = \frac{1}{6}$ & $P(X=1) = \frac{1}{6}$

$$\therefore P(X=4,\ X=1) = \frac{5!}{2! \cdot 3!} \cdot \left(\frac{1}{6}\right)^2 \cdot \left(\frac{1}{6}\right)^3$$

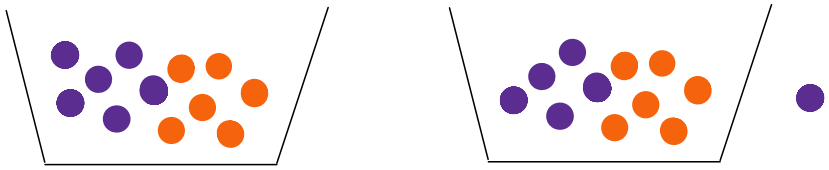
$$= \frac{5 \times 4 \times 3!}{2 \times 1 \times 3!} \times \frac{1}{36} \times \frac{1}{216}$$

$$= \frac{10}{36 \times 216} = 0.1\%$$

(5) Hypergeometric Distributions :-

## ⑤ Hypergeometric Distributions :

It is used when we want to select items without replacement from a finite set.



I. P(Blue) = 0.5 II. P(Blue) = 0.45

Formula →

$$P(X=x) = \frac{{}^{K}C_{x} \cdot {}^{N-K}C_{n-x}}{{}^{N}C_{n}}$$

where,

N = population size
K = Total success items
n = no. of draws
x = No. of successes obtained

Ques. A bag contains ⇝ (1) 6 blue balls
(2) 6 orange balls

Randomly draw 4 balls without replacement
What is the probability of getting exactly 2 blue balls?

Sol.

given :- N = 12
n = 4
K = 6
x = 2

$$P(X=2) = \frac{{}^{6}C_{2} \cdot {}^{12-6}C_{4-2}}{{}^{12}C_{4}} = \frac{{}^{6}C_{2} \cdot {}^{6}C_{2}}{{}^{12}C_{4}}$$

$$P(X=2) = \frac{{}^{2}C_{2} \cdot {}^{4-2}C}{{}^{12}C_{4}} = \frac{{}^{2}C_{2} \cdot {}^{2}C_{2}}{{}^{12}C_{4}}$$

$$P(X=2) = 0.45 \text{ or } 45\%$$

## ★ Characteristics of Discrete Distributions :-

### ① Binomial Dist :-

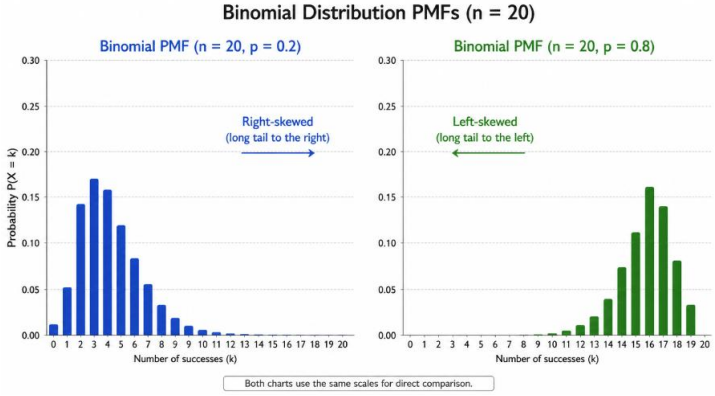
Mean ($\mu$) = $np$

where, $n$ = no. of trials

$p$ = prob. of success

Variance ($\sigma^2$) = $np(1-p)$

Shape :-

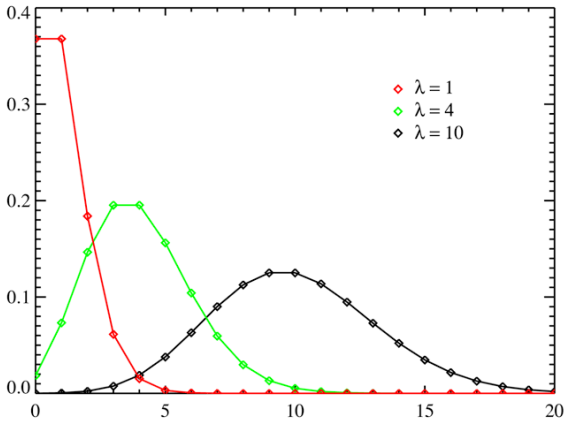



### ② Poisson's Dist :-

Mean ($\mu$) = $\lambda$

Variance ($\sigma^2$) = $\lambda$

Shape :-




when $\lambda$ is small → right skewed dist., variance is small

" " is big → symmetric curve, variance inc.

(3) Multinomial Dist. :-

Mean ($\mu$) = for $i^{th}$ outcome → $\mu_i = np_i$

Variance ($\sigma^2$) = for $i^{th}$ outcome → $np_i(1-p_i)$

Covariance =>

$$Cov(X_i, X_j) = -n \cdot p_i \cdot p_j$$